

TradeMind Master Blueprint

One document for the whole product: what TradeMind is, how its brain thinks and remembers, what happens when it's confused or its memory fails, how data flows technically, what exists in SwingTradeML today, every requirement, and the plan to build it. It follows the structure of your "Complete Architecture, Requirements & Product Blueprint" PDF and fills every section with the real code, the agreed requirements and the review's recommendations.

DATE 29 Sep 2026	PRODUCT TradeMind on SwingTradeML	STATUS Draft for approval
OWNER Naresh Gudiganti		

How to read. Part A explains the product and business. Part B is the brain: mind diagram, memory, confusion and failure handling, decisions, contracts and safety. Part C is the technical design. Part D lists everything the application has today, every requirement, the implementation plan and the open problems. Part E reviews your two newest pictures, the Central Brain and the Experience and Application architecture, and lists what they add to the requirements. Record disagreements in the problem log (chapter 35) rather than changing contracts silently.

- Built in code
- Partly built
- Planned
- A

 built
- B

 extend
- C

 new
- D

 future
- changed from your picture

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01 Executive understanding

TradeMind is not only a stock scanner, a signal generator, one ML model or a trading bot. It is a **central intelligence brain** that receives market, portfolio, user, news, macro, historical and system information; builds context; recognises situations; reasons over evidence; applies risk and constitutional safety rules; decides; tracks what actually happened; and learns from outcomes. A signal is only one of its outputs.

WHERE WE ARE	TODAY (29 SEP 2026)
What runs	SwingTradeML: a personal system on one DigitalOcean droplet. Nightly scan at 15:45, 53 features, three cap-tier LightGBM models, 13 risk rules, exits every 60 seconds, paper trading by default behind a two-flag live latch.
Evidence	Model AUC 0.556–0.583. Top 5% of picks: +0.22 to +0.28R per trade in a simple test. Full-system replay on ₹10 lakh: –0.1% to –0.4% per 8-month window after costs, while NIFTY made +6%.
Plans	Free / Plus / Pro gating exists on branch feature/plans-manager (not merged; switch off by default; no payments).
Gates before a public product	1 · a measured edge after costs. 2 · separate user accounts and data. 3 · SEBI Research Analyst registration and licensed data for users.

The plan in one line: keep the working engine, wrap it in stable contracts, add intelligence one plug-in at a time behind the safety layer, and measure every step against today's baseline.

02 Principles

The blueprint's ten principles, merged with the safety principles from the review. When two conflict, the lower number wins.

#	PRINCIPLE	IN PRACTICE	TODAY
1	Safety rules never change	Stop, risk per trade, caps, drawdown halt are locked for every plug-in, plan and setting	Yes: per-strategy risk overrides refused (commit 6a7e76e)
2	No uncontrolled changes	Models, rules, playbooks change only through tests, shadow runs and owner approval	Partly: models need activation; approvals inbox planned
3	Unknown is a valid state	The brain can say WAIT or UNKNOWN instead of forcing a call	Partly: HOLD with reasons; UNKNOWN planned
4	No silent failures	Detect → report → recover → learn; stale data lowers quality and can stop a decision	Partly: feed alerts, heartbeat, deploy verify
5	Every important decision is reproducible	Decision ledger with inputs, versions and times; replay on demand	Partly: signals stored with features and reasons
6	Evidence over a single score	Evidence for and against, probabilities with track record, never promises	Partly: reasons and calibration report
7	Intelligence is not one ML model	Rules, setups, several models, memory and context together	Partly: rules + ML today
8	Brain independent of infrastructure	Ports and adapters; external systems are replaceable	Partly: Broker interface already a port
9	Cache is never the source of truth	PostgreSQL holds truth; Redis only speeds things up	Yes: Redis holds only the heartbeat
10	New capabilities are additive, versioned and reversible	Feature flags, shadow mode, rollbacks; learn from outcomes, not from every user click	Partly: rollbacks, shadow strategies, plans switch

03 The architecture: your picture and the final design

Figure 3.1 · Your latest picture TradeMind Complete Trading Intelligence Platform

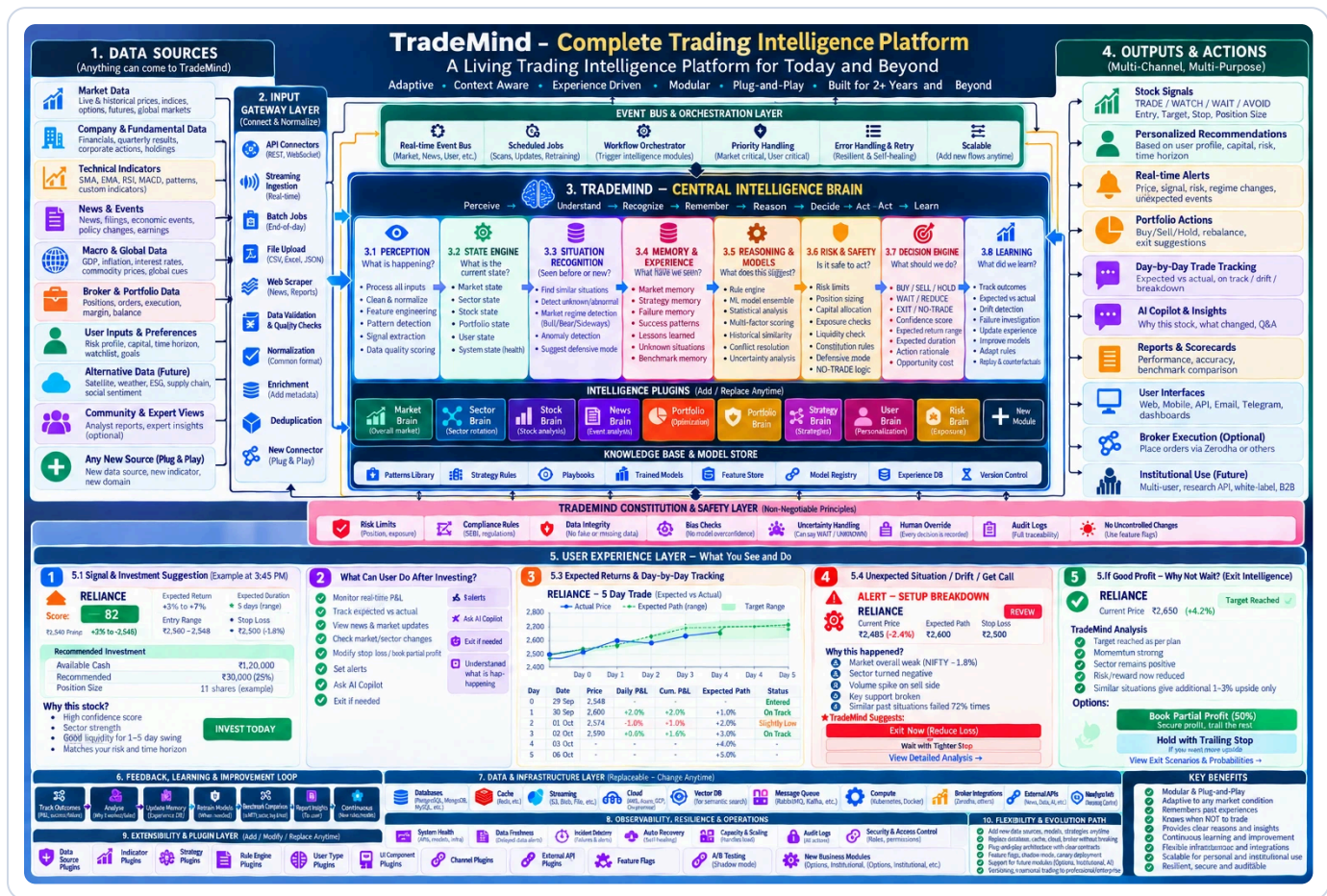


Figure 3.2 · Final architecture, with the review's fixes applied items marked ✎ differ from your picture

1 Data sources Licensed, stored by date, tested before use

- Market data
- Company and fundamentals
- Technical indicators
- News and filings (NSE/BSE)
- Macro calendar and FII/DII ✎
- Broker and portfolio
- User inputs and preferences
- New source (plug-in)



2 Input gateway Nothing reaches the brain unchecked

- Connectors: API, WebSocket, file ✎
- Validation and quality
- Normalisation
- Enrichment
- De-duplication
- Versioning by time and source ✎
- Security and access



3 Common data contracts ✎ Nine stable, versioned contracts

- Decision
- Evidence
- MarketContext
- PortfolioContext
- RiskContext
- OrderIntent
- Outcome
- Experience
- SystemHealth



4 Event bus and orchestration

Steps start when inputs are ready

- ☒ Scheduled jobs
- ☐ Redis event queue
- ☒ Dependency-driven workflow
- ☐ Priority: exits first
- ☒ Retry and dead-letter
- ☐ Replay



5 Central intelligence brain

Perceive → state → recognise → remember → reason → check safety → decide → learn



PLUG-INS

- ☒ Market
- ☒ Sector and theme
- ☐ Stock (DNA)
- ☐ News
- ☐ Portfolio (incl. optimisation)
- ☒ Strategy
- ☒ Risk
- ☐ User
- ☐ New module

KNOWLEDGE BASE AND MODEL STORE

- ☐ Patterns
- ☒ Rules
- ☐ Playbooks
- ☒ Trained models
- ☐ Feature store
- ☒ Model registry
- ☒ Experience DB
- ☒ Versions



6 Constitution and safety layer

Non-negotiable, versioned; has the final word

- ☒ Locked risk limits
- ☐ SEBI compliance
- ☒ Data integrity
- ☒ Bias checks
- ☐ Can say WAIT / UNKNOWN
- ☒ Human override: stop or overrule
- ☒ Audit logs
- ☒ No uncontrolled changes



7 Decisions

Three small vocabularies



8 Outputs by plan

Same decisions; each plan sees its share

- ☒ Signals with trigger, levels, amount
- ☐ Holding actions
- ☐ Market banner
- ☒ Alerts
- ☒ Day-by-day tracking
- ☐ Copilot (Pro)
- ☒ Reports and scorecards
- ☐ Order-list export (Pro)
- ☐ Broker execution, institutional: later



9 Foundation Learning loop · observability · small infrastructure · governance and plug-ins

LEARNING LOOP

● Track outcomes

○ Analyse why

○ Propose changes ✎

● Owner approval

● Shadow, then promote

GOVERNANCE AND PLUG-INS

● Versioning

● Feature flags

○ Change registry

● Rollbacks

● Plug-ins

OBSERVABILITY

● Health

● Freshness

○ Incidents

● Self-healing, infra only ✎

● Security

INFRASTRUCTURE ✎

● PostgreSQL + TimescaleDB

● Redis

○ Object storage

● Docker on DigitalOcean

● Kite via Broker port

04 Business flows

Figure 4.1 · A user's journey visitor → paying user

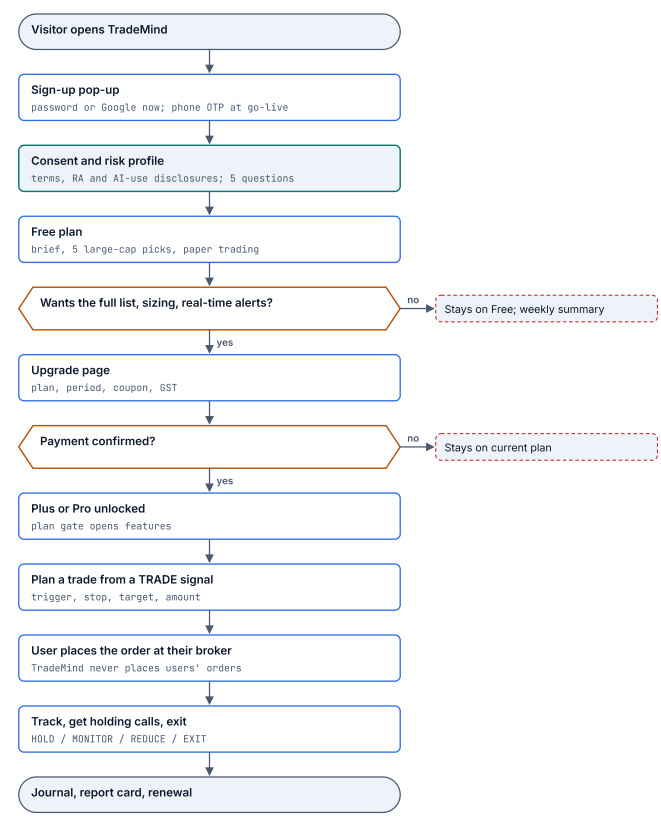
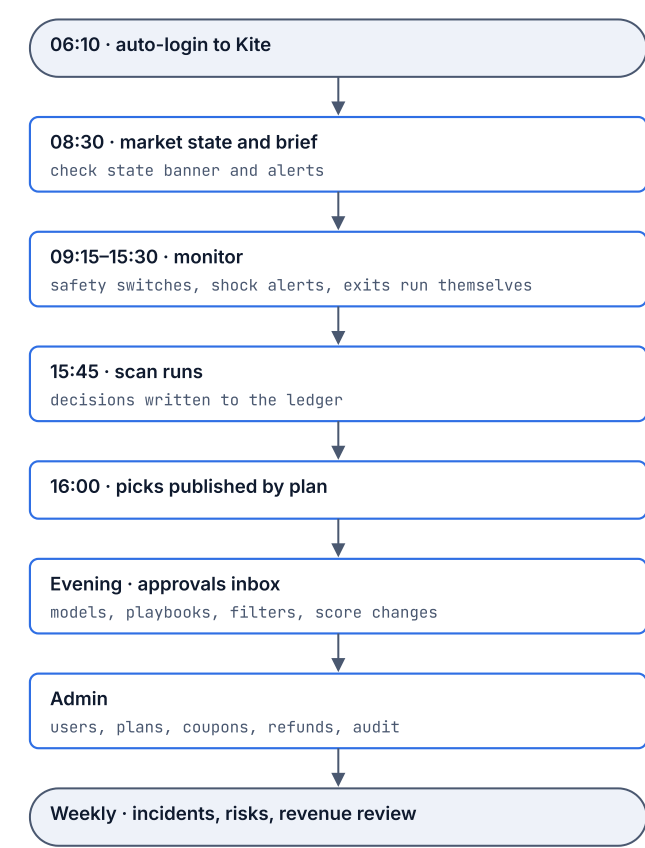


Figure 4.2 · The owner's day operating the product



BUSINESS ELEMENT	DECISION SO FAR
Plans	Free (Discover), Plus (Decide), Pro (TradeMind); one brain, three views
Pricing	About 50% below comparable products; Plus about half of Pro; benchmark seen: Tickertape Pro ₹399 a month, ₹899 for 3 months, ₹2,399.2 a year
Launch offer	6 months of Pro free, if the SEBI consultant confirms it
Billing periods	1 and 3 months until the RA advance-fee limit is confirmed
Sign-in	Password or Google now; phone OTP and Okta at go-live
Orders	Users place their own orders at their broker; the bot trades only the owner's book

05 Free, Plus and Pro

AREA	FREE · DISCOVER	PLUS · DECIDE	PRO · TRADEMIND
Stock signals	5 large-cap, state and top reason	All, trigger, levels, Expected R, amount	+ setup odds, horizons, model agreement
Holding actions	—	HOLD / MONITOR / REDUCE / EXIT	+ exit scenarios, opportunity cost
Market state	Banner	+ index and sector trends	+ closest past episode, shock centre
Alerts	End of day, in app	Real time: push, Telegram	+ WhatsApp, event shocks
Day-by-day tracking	Paper trades	Your trades	+ typical range, drift reasons
Reports	Verified summary	Score report card	+ filter, factor, model scorecards
Portfolio	—	Health score, 1% sizing	+ correlation, factors, what-if, custom risk
Models, strategies	Default safety model	1 model, 1 strategy	Any; agreement mode; presets
AI	Short brief	Full brief	Personal brief, copilot, research lab
Orders	Paper trading	Read-only broker link	+ order-list export

06 The five user moments

Your five moments, rebuilt so every number follows today's engine: stop -4% and target +8% from entry, half booked at +5%, a trailing stop on the rest, 1% risk per trade, and position caps from the capital ladder (25% per position at ₹1 lakh, 10% at ₹10 lakh, interpolated). Probabilities are illustrative. 

1 · SIGNAL · 15:45

Illustrative stock "ABC"

ABC · Large cap · Pullback

TRADE

Trigger (next session)

above ₹1,002.50

Stop · first target · target

₹963 · ₹1,053 · ₹1,083

Holding period

up to 15 trading days

Reached +5% first (similar signals)

58% of 120

Expected R after costs

+0.3R

Your capital

₹1,20,000

Suggested amount (cap ≈ 24%)

₹28,500

Quantity at ₹1,003

28 shares (₹28,084)

At risk if the stop hits

₹1,120 (0.93%)

Plan this trade

Add to watchlist

Why this stock?

2 · AFTER INVESTING

Daily

Morning brief

08:45

Live P&L and drift status

all day

Alerts: stop, target, event, regime, shock

by plan

Ask the copilot

Pro

What changed today

16:15

ON TRACK

SLIGHTLY OFF

WARNING

BREAKDOWN

Set alert

Tighten stop

Book partial profit

Exit at my broker

The app's stop only ever moves up.

3 · DAY-BY-DAY

Illustrative

DAY	PRICE	P&L	TYPICAL RANGE	STATUS
0	₹1,003	0.0%	—	Entered
1	₹1,015	+1.2%	+0.2 to +1.8%	On track
2	₹1,009	+0.6%	+0.5 to +2.8%	Slightly off
3	₹1,030	+2.7%	+0.9 to +3.6%	On track

Typical range = middle half of similar past trades on that day; shown only with 20+ similar trades.

4 · BREAKDOWN · DAY 4

Illustrative

ABC ₹975 (-2.8%) · stop ₹963 not hit

REDUCE

Sector weak; NIFTY -1.8%; selling volume spike

reasons

Similar situations that later hit the stop

64% of 45

Exit now (-2.8%)

Tighten stop to ₹968

Keep stop ₹963

At ₹963 the stop exits automatically: "Stop hit, exited at ₹962". A stop is never loosened.

ABC ₹1,053 (+5%)

HOLD rest

Half booked automatically at +5% **14 of 28 shares**

Trailing stop on the other 14 **moves up only**

Went on from +5% to +8% **47% of 88**
(similar)

Hold with trailing stop

Exit the rest

Exit scenarios

07 User problem → architecture

USER PROBLEM	BRAIN PARTS THAT ANSWER IT	REQUIREMENT ITEMS
1 · Signal arrives late, cash is available	Capital deployment, opportunity cost, trigger entry	F06, F14, F15, F40, A2
2 · What do I do after investing?	Position tracker, monitoring, action centre	F12, F21, F52, P5
3 · Expected vs actual, day by day	Expectation engine and drift	F50, F05
4 · The thesis breaks	Re-evaluation, safety gate, holding call	F11, F12, F13, F25
5 · Profit is already good	Exit intelligence, opportunity cost	A6, F13, F51, F14
6 · Bearish or unknown market, signals failing	Situation recognition, experience memory, defensive mode	F35, F36, F39, C8
7 · System or infrastructure failure	Health engine, incident memory, safe degradation	F24, C6, C9, P7

08 The TradeMind mind

The brain drawn as a mind: what it senses, how it remembers, where it reasons, what stops it from acting when it's confused or afraid, how its conscience (the constitution) has the final word, and how it learns overnight. Each region is a real part of the design, named in the table below.



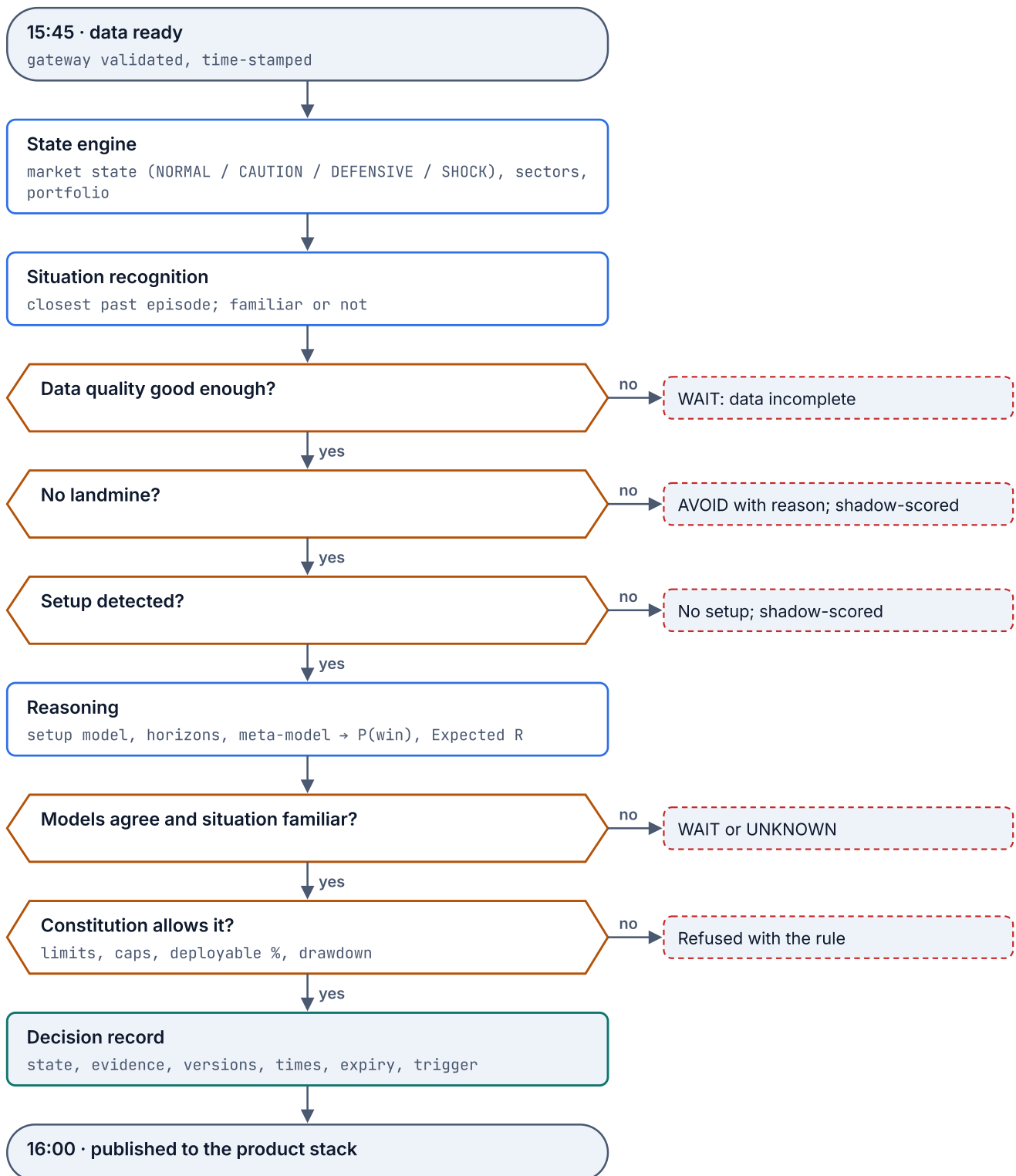
REGION	LIKE IN A PERSON	IN TRADEMIND	IN CODE TODAY
Senses	Eyes, ears, touch	Input gateway and data contracts	Kite quotes and candles, NSE feeds, holdings sync
Perception	Making sense of what you see	Validation, normalisation, time-stamps, data-quality score	Feed checks, spike guard, 53 level-invariant features
Working memory	What's in your head right now	State engine	Regime (deployable.py), positions, cash
Recognition	"I've been here before"	Situation recognition	Regime only; market memory planned (F35)
Long-term memory	Experience	Eight memory stores (chapter 10)	Signals, trades, predictions; others planned
Alarm	Fear response	Shock detector → market state	Regime table, drawdown halt; shock detector planned
Vital signs	Heartbeat, breathing	Health engine	Heartbeat, check_kite_login, deploy verification

REGION	LIKE IN A PERSON	IN TRADEMIND	IN CODE TODAY
Reasoning	Thinking it through	Rules, setup models, meta-model, Expected R	ml_swing thresholds and filters
Conflict monitor	"Something doesn't add up"	Uncertainty and agreement → WAIT / UNKNOWN	Planned (F25)
Conscience	Values you won't break	Constitution and safety layer	risk.check_entry, locked risk fields, kill switch, live latch
Decision	Choosing	Decision engine and contract	SignalDecision → Signal rows
Hands and voice	Acting and speaking	Broker adapter (owner only), alerts, briefs, copilot	Paper/Kite brokers, Telegram
Learning	Reflecting on results	Outcome attribution, lessons, proposals with approval	evaluate_signals, calibration
Sleep	Memory consolidation at night	Evening jobs: scoring, scorecards, backups	16:15–16:17 scoring jobs; backups on deploy

09 Brain modules

MODULE	TAKES IN	GIVES OUT	TODAY	ITEMS
1 Perception	Raw inputs	Clean, dated, quality-scored data	Feeds, features, spike guard	F24, C3
2 State engine	Perceived data	Market, sector, stock, portfolio, user, system state	Regime, positions	F01, F02, F46, C8
3 Situation recognition	State + memory	Familiar / unfamiliar / abnormal; closest past episode; regime transition	Regime only	F26, F35
4 Memory and experience	Decisions, outcomes, incidents	Relevant experience for this situation	Signals, trades, predictions	F18, F31, F35, C13
5 Reasoning	Features, setups, context, memory	Probabilities, Expected R, evidence for and against	One classifier per cap tier + filters	F04–F07, F22
6 Risk and safety	Candidate decision + portfolio + policy	Allowed or refused, size, reason	13 rules, ladder, deployable %	A2, C7
7 Decision engine	Everything above	One state per stock + contract	BUY / HOLD / EXIT	F10, F25, C2
8 Learning engine	Outcomes vs expectations	Scorecards, lessons, proposals	Scoring, calibration	F23, F31, F37–F39

Figure 9.1 · How the modules run each evening trading core · 15:45



10 Memory architecture

Memory is split by **meaning**, not by storage product. PostgreSQL holds the truth; Redis only speeds things up; object storage keeps large raw files; similarity search can later use pgvector inside PostgreSQL.

MEMORY	HOLDS	WRITTEN BY	READ BY	STORED IN	TODAY
Market experience	Past episodes: corrections, bear phases, shocks; how they unfolded	Market memory job	Recognition, bear mode	market_episodes, episode_replays	Planned
Strategy experience	Results by strategy, setup and regime	Scorecards	Strategy brain, meta-model	score_band_stats, scorecard tables	Partly
Failures	Why each loss happened	Failure investigator	Lessons, DNA	trade_journal.failure_label	Planned
Unknown situations	Times the brain said UNKNOWN and what followed	Decision engine	Recognition	decision_records + outcomes	Planned
Decision ledger	Every decision with inputs, versions, times	Decision engine	Replay, copilot, audit	signals today → decision_records	Partly
Outcomes	What happened after each decision	Evening scoring	Learning	signals.outcome, trades, predictions	Built
User preferences	Profile, choices, taken and skipped signals	User actions	Ranking, sizing	users (+ plan), journal	Partly
System incidents	Failures, cause, fix, prevention	Health engine	Operations, deploys	incidents (new)	Planned

Time truth: five timestamps on every piece of data

TIME	MEANING	WHY
Event time	When it happened in the market	The fact itself
Received time	When we fetched it	Feeds are already stored dated on fetch
Processed time	When we cleaned and stored it	Debugging lag
Available-to-model time	From when a model may use it	Stops look-ahead bias in backtests
Decision time	When the brain used it	Replay exactly what was known

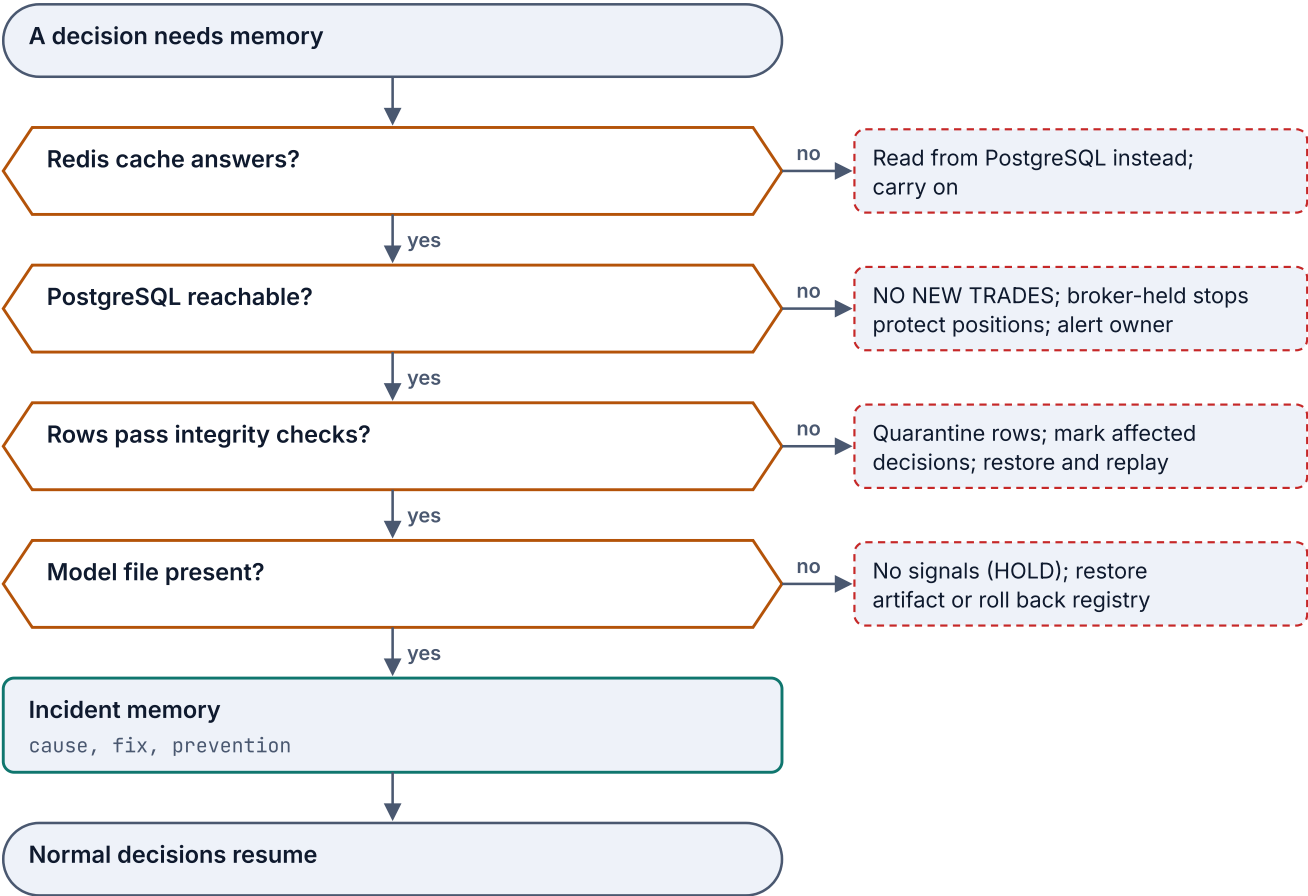
11 When the mind struggles: confusion, fear, memory crash

A good trader knows when they're confused, afraid or not thinking clearly, and doesn't trade then. TradeMind has the same instincts built in. Each condition below has a detection rule, a safe behaviour, a recovery path and something it remembers.

CONDITION	LIKE A PERSON WHO...	HOW IT'S DETECTED	WHAT TRADEMIND DOES	RECOVERY AND MEMORY
Confusion	hears contradicting advice	Models disagree (below 4 of 6); evidence for and against both strong	WAIT with the reason; no size given	Re-checked next scan; kept in the decision ledger
Unfamiliar situation	has never been here before	No close match in market memory	UNKNOWN; smaller limits; defensive if market-wide	Stored in unknown-situation memory with its outcome
Fear / shock	sees a sudden crash	NIFTY sharp fall, VIX spike, breadth collapse, shock on high volume	SHOCK state: no new trades; holdings reviewed; stops unchanged	Back to normal after 2 calm days; episode recorded
Overconfidence	believes their own hype	Score band's real hit rate 10+ points below stated	Band marked "under review"; stops showing as TRADE	Recalibration proposed; owner approves
Blurred senses	can't see clearly	Stale or missing feed; failed validation	Data-quality score drops; affected decisions become WAIT	Feed alert; retry; incident logged
Short-term memory loss	forgets what they were doing	Redis cache empty or down	Reload from PostgreSQL; no decision from cache alone	Automatic; nothing lost
Long-term memory loss	loses their memory	Database unreachable	NO NEW TRADES; exits rely on broker-held stops (GTT, planned)	Restore from backup (database dump on every deploy); incident logged
Corrupted memory	remembers something wrong	Integrity checks, spike guard, restated data, look-ahead tests	Quarantine the bad rows; mark decisions using them	Restore and replay affected decisions
Amnesia after a change	wakes up without skills	Model file missing (backup job alerts), wrong model version	No model = no signals (HOLD); never guess a model	Restore artifact; registry rollback in one call
Split brain	thinks they own something they sold	App's positions $\neq$ broker's holdings	Broker is the truth for holdings; alerts stop for sold stocks	holdings_sync and reconcile_orders fix it; incident logged
Exhaustion	is overloaded	Scan too slow; job queue backlog	Exits first, scans second, Pro research last	Scale workers; capacity note in incidents

CONDITION	LIKE A PERSON WHO...	HOW IT'S DETECTED	WHAT TRADEMIND DOES	RECOVERY AND MEMORY
Coma	stops responding	Worker heartbeat missing	Alert the owner; deploy verification fails	Restart; release rollback to previous images
Imagining things	makes up facts	Copilot answer without a citation	Answer withheld; says what it doesn't know	Evaluation set updated

Figure 11.1 · Memory-crash recovery path cache → database → backup



12 Decision vocabulary and market states

Your picture and the blueprint used different words (BUY / SELL / HOLD / WAIT / REDUCE / EXIT / NO-TRADE / UNKNOWN; TRADE / WATCH / WAIT / AVOID). The final design uses three small sets, one per situation. UNKNOWN is its own state, as the blueprint asks. 📌

SITUATION	STATE	MEANING	BLUEPRINT WORD
A stock you don't own			
	TRADE	All gates pass, positive Expected R, trigger set for next session	BUY
	WATCH	Good stock, entry not right yet	—
	WAIT	Models disagree, data poor, event due, regime changing	WAIT
	UNKNOWN	Situation outside experience; not enough evidence	UNKNOWN / INSUFFICIENT EVIDENCE
	AVOID	Landmine, negative Expected R or poor risk	NO-TRADE (one stock)
A stock you hold			
	HOLD	Reason still holds	HOLD
	MONITOR	Reason weakening	—
	REDUCE	Book part or tighten the stop	REDUCE
	EXIT	Leave the position	SELL, EXIT
The whole market (maps today's six regimes)			
	NORMAL	Regime normal, strong or very strong	NORMAL
	CAUTION	Regime weak, or unknown (too little index history)	CAUTION
	DEFENSIVE	Regime stressed: deploy $\leq 10\%$, large caps, tested setups only	DEFENSIVE
	SHOCK	Sudden crash signs: no new trades; exits continue	SHOCK / EXTREME-SHOCK, NO-TRADE

"SELL" is dropped: trading is delivery (CNC) only, with no short selling.

13 Decision contract and stable contracts

Every decision uses one versioned contract. It separates the brain from screens, brokers and infrastructure. Today's `SignalDecision` is the seed.

FIELD	MEANING	TODAY
<code>decision_id</code>	Unique id	Partly · <code>signals.id</code>
<code>timestamp</code> · decision time	When decided	Partly · <code>signals.created_at</code>
<code>instrument</code>	The stock	Built · <code>instrument_id</code>
<code>action</code>	State from chapter 12	Partly · BUY / HOLD / EXIT
<code>confidence</code>	Calibrated probability	Partly · <code>confidence</code>
<code>system_confidence</code>	How sure the whole system is	New · –
<code>data_confidence</code>	Data-quality score	New · –
<code>expected_horizon</code>	Holding period	Built · <code>horizon_days</code>
<code>expected_return_range</code>	Range with probabilities	New · –
<code>risk / stop</code> reference	Stop and target	Built · <code>stop_loss</code> , <code>take_profit</code>
<code>position_size</code> suggestion	Amount and quantity	Built · <code>RiskDecision.quantity</code>
<code>evidence list</code> , reasons, conflicts	For and against	Partly · <code>reason</code> , <code>features</code> , <code>rejection_reason</code>
<code>market_state</code> , <code>portfolio_state</code>	Context at decision time	New · –
<code>model</code> , <code>strategy</code> , <code>policy</code> versions	Traceability	Partly · <code>strategy_id</code> ; model by name
<code>expiry</code>	When the decision lapses	New · –
<code>audit</code> references	Links to audit rows	Partly · <code>risk_events</code>

The nine stable contracts

CONTRACT	CARRIES	EXISTS TODAY AS	STATUS
Decision	State, confidence, expectation, size, evidence, versions, expiry	<code>SignalDecision</code> + <code>signals</code> row	Partly
Evidence	Reasons for and against with sources	<code>Signal.features</code> , <code>reason</code> text	Partly
MarketContext	Market state, regime, trends, shocks	<code>deployable.current_deployable()</code> , <code>market_context</code>	Partly
PortfolioContext	Holdings, cash, exposures, drawdown	<code>risk.open_holdings()</code> , <code>portfolio_value_and_cash()</code>	Partly
RiskContext	Limits, budget, allowed tiers	<code>limits_for()</code> , <code>RiskDecision</code>	Partly

CONTRACT	CARRIES	EXISTS TODAY AS	STATUS
OrderIntent	What to buy or sell, how, limits	OrderRequest (brokers/base.py)	Built
Outcome	What happened after a decision	Signal.outcome, Trade, OrderResult	Partly
Experience	A remembered situation and its lesson	—	New
SystemHealth	Freshness, services, models, broker link	Heartbeat, system_state	Partly

14 Constitution and safety layer

Non-negotiable and versioned. It surrounds decision-making: every decision passes it last, and it decides when the system must refuse to act confidently.

RULE	VALUE TODAY	WHERE IN CODE	STATUS
Risk per trade	1% of account	RISK_PER_TRADE_PCT, risk.calculate_quantity	Locked
Stop and target	−4% and +8% from entry; half at +5%	ML_STOP_RETURN_PCT, ML_TARGET_RETURN_PCT, ML_FIRST_TARGET_PCT	Locked
Drawdown halt	No new entries at 15%	MAX_PORTFOLIO_DRAWDOWN_PCT, check_entry	Locked
Position and sector caps	Capital ladder ₹10K → ₹1Cr	services/limits.py	Locked
Time stop	30 calendar days	exit_policy.py	Locked
Live trading latch	Two flags, checked twice	brokers/__init__.get_broker, KiteBroker.place_order	Built
Emergency stop and human override	Kill switch, exits switch	system_state, Safety page	Built
No per-strategy risk overrides	Refused	commit 6a7e76e	Built
Landmines	Results 3 days, corporate actions 2 days, ASM/GSM/T2T	services/avoid.py	Built
Minimum evidence and refusal rules	WAIT / UNKNOWN when below	—	New (C7)
Stale-data handling	Unavailable, never silently old	Feed alerts; quality score planned	Partly
Model and version traceability	Model, strategy, policy version on every decision	—	New (C2)
Compliance	RA disclosures, advertising code, AI-use disclosure	—	Planned (P3)
Constitution version	Rules stored as a versioned policy	Code constants today	New (C7)

15 Expectation, re-evaluation and exits

Expectation envelope on every TRADE

- Holding horizon: up to 15 trading days, with odds at 3 / 5 / 10 / 15
- Return range and probabilities, each with sample size
- Stop and target reference; trigger price and expiry
- Typical path of similar past trades, as a band

Drift status after entry

ON TRACK

SLIGHTLY OFF

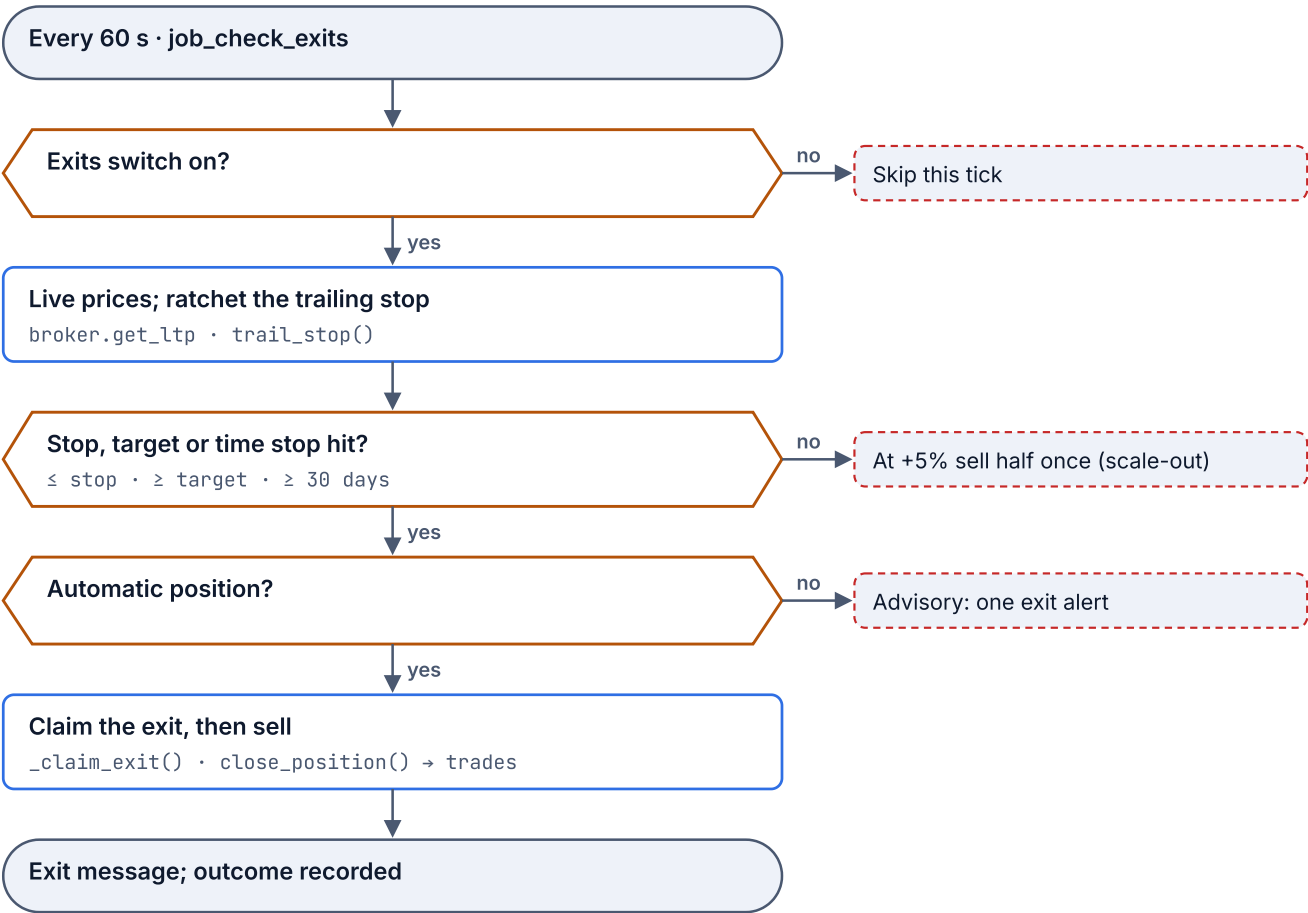
WARNING

BREAKDOWN

Re-evaluation triggers

Regime change	F26
Signal deterioration	F11
News or global shock	F08, F09
Sector breakdown	F02
Volume anomaly	F09
Expectation drift	F50
Model disagreement	F25
Stop proximity	A6
Portfolio risk	F15, F16
System health drop	F24
Better opportunity	F14

Figure 15.1 · Exit loop today (every 60 seconds) `services/execution.check_exits`



16 Capital, opportunity cost and defensive mode

ACCOUNT (LADDER RUNG)	MAX POSITIONS	PER STOCK	REGIME TODAY	DEPLOYABLE	MARKET STATE
₹10,000	2	50%	very strong	85%	NORMAL
₹1,00,000	4	25%	strong	75%	NORMAL
₹10,00,000	8	10%	normal	55%	NORMAL
Values between rungs interpolate on log(account value); up to ₹1 crore.			weak	30%	CAUTION
			stressed	10%	DEFENSIVE
			unknown	30%	CAUTION
			sudden crash (new)	0% new	SHOCK

Capital allocation is a decision layer: the same signal can get a different amount depending on market state, portfolio state and the user's policy. Opportunity cost compares each holding's remaining edge with new candidates and shows a review when the gap is 1R or more; it never sells automatically.

17 Technical flows

Figure 17.1 • Entry path today: from a BUY decision to a position `execution.process_decision` • `risk.check_entry` • `brokers`

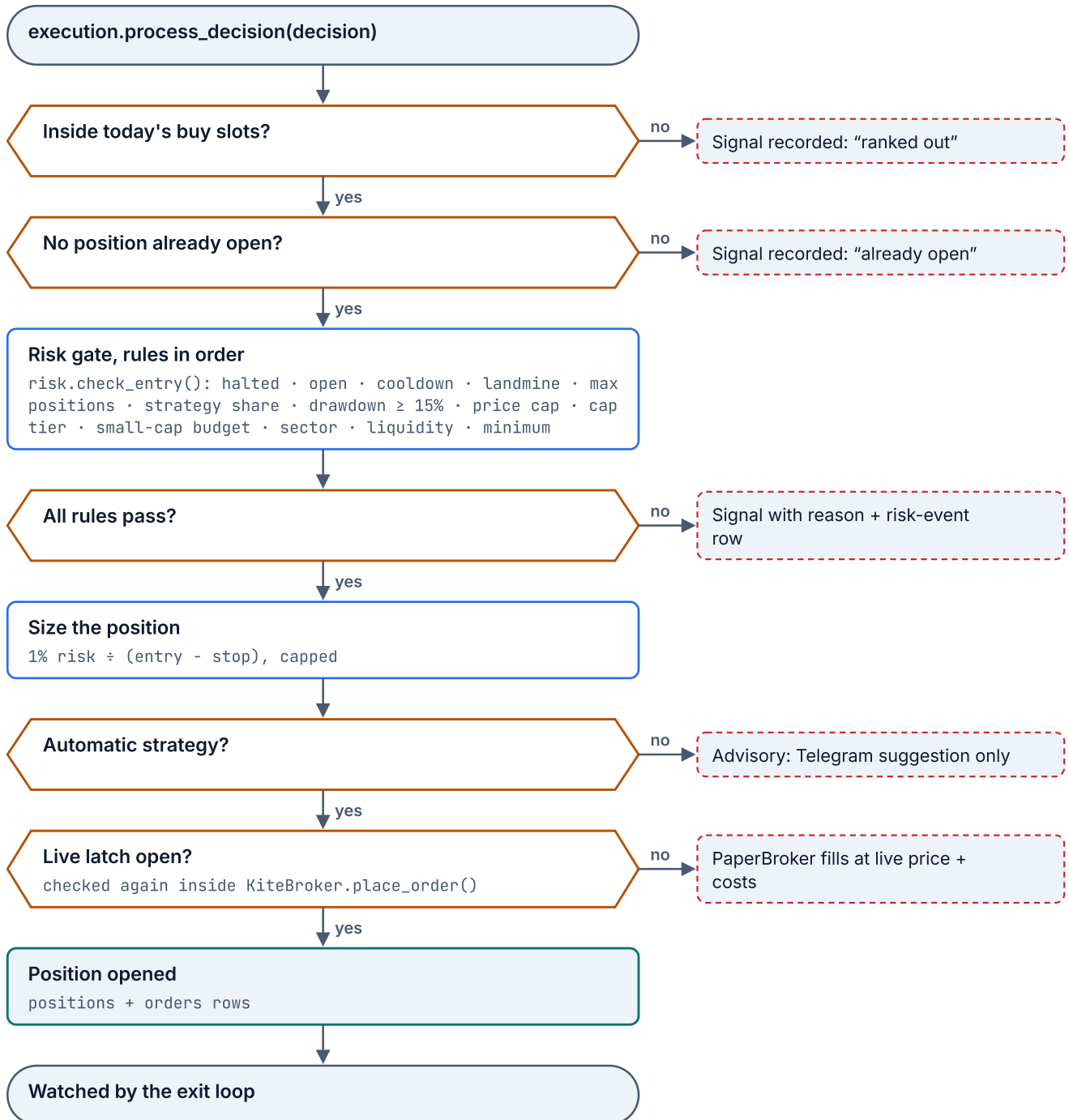
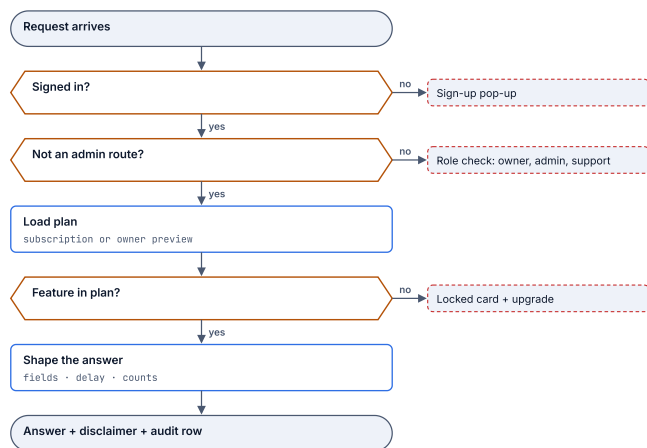


Figure 17.2 • Every request through the plan

Figure 17.3 • Learning loop `scorecards` •

gate api/deps.py



lessons · approvals

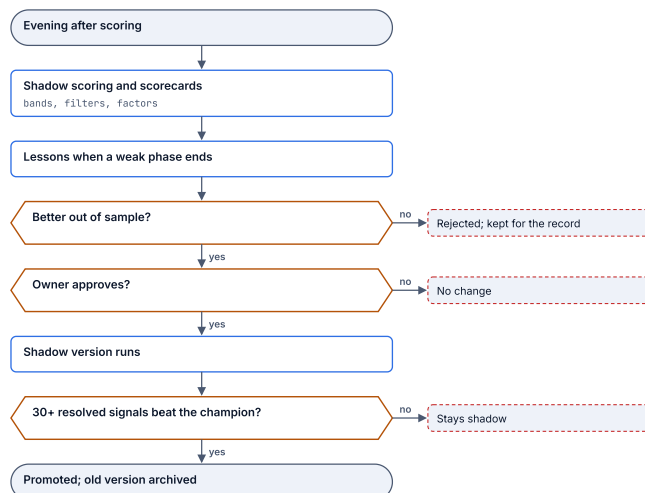
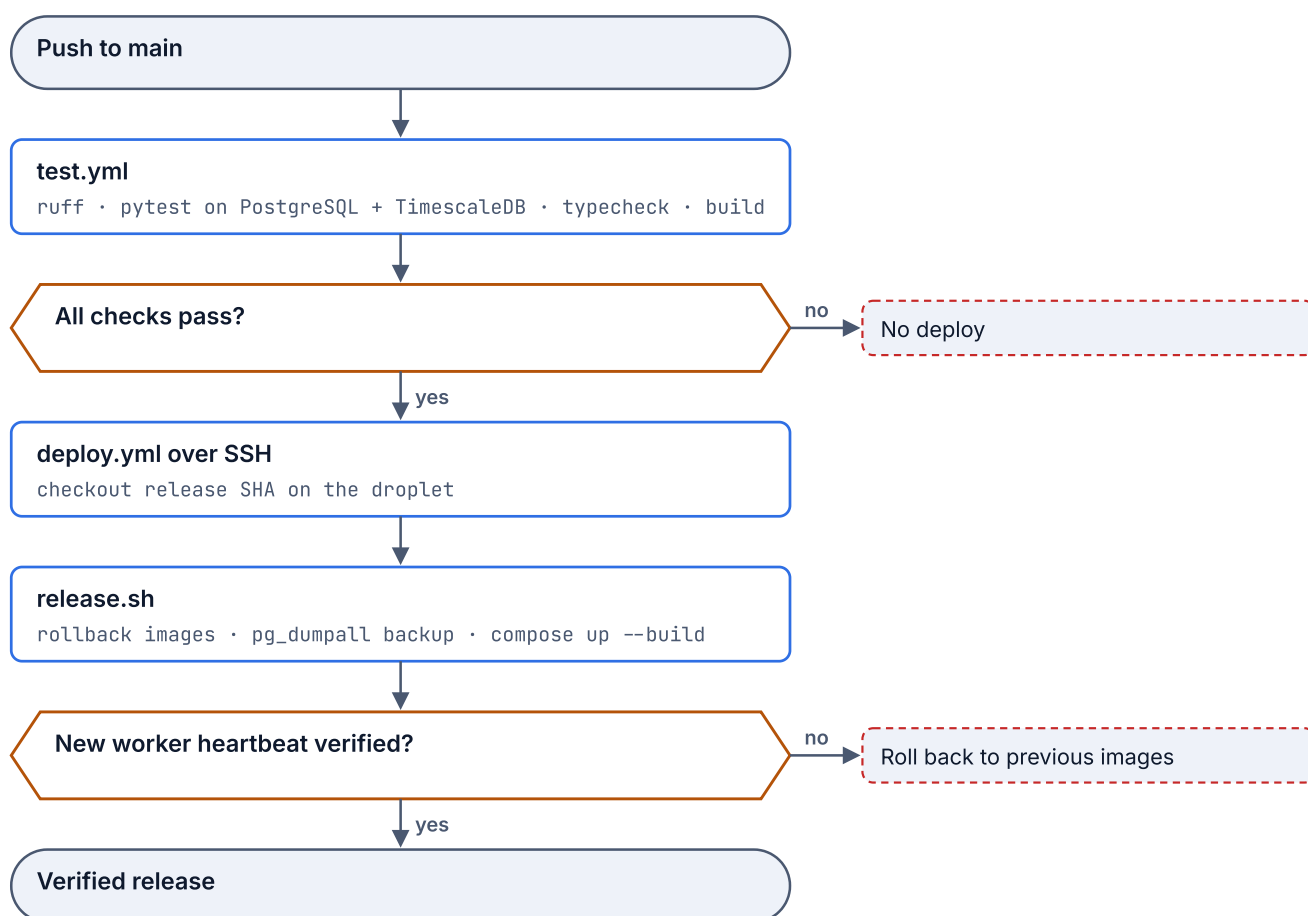


Figure 17.4 · Deploy path today `test.yml` → `deploy.yml` → `release.sh`



18 Data and event architecture

SOURCE	ADAPTER TODAY	SCHEDULE	STORED IN	USERS MAY SEE
Kite candles and quotes	services/ingestion.py, brokers/kite.py	15:40 daily; quotes every 60 s	candles (hypertable), quotes	Needs licensed vendor
NSE delivery, deals, FII/DII, restriction lists, events	services/market_feeds.py	08:30, 19:20, 22:05	feed tables, dated on fetch	Derived summaries
Instrument master and index lists	CLI sync-instruments, sync-index	Sunday 08:00	instruments	Yes
Holdings and fills (owner)	holdings_sync.py, fills.py	15:48; 15:50 and 19:30	positions, broker_fills	Owner only
Filing text, global cues, licensed feed	New adapters	Planned	event_items, market_state_daily	Yes, by plan

The input gateway checks timestamp, freshness, schema, provenance and quality before anything reaches the brain. Event orchestration adds a Redis-backed queue, dependency-driven steps (a job starts when its inputs are ready, not at a fixed clock time), priorities (exits before scans), retries with a dead-letter list, and replay.

19 Models and strategies

LIFECYCLE STAGE	TODAY	TARGET
Research and training	CLI swingtrade train, purged 5-fold walk-forward	Research lab jobs
Backtest	services/backtest.py with real costs	+ episode replay, bias checks
Paper	PaperBroker, same pipeline	Same
Shadow	Advisory strategies (barrier models)	Shadow versions via system versioning
Canary and production	Registry: TRAINED → ACTIVE, previous ARCHIVED	Promotion after 30+ resolved signals beat the champion, with approval
Monitoring and retirement	Calibration, signal evidence	Drift monitor, scorecards, retirement rule

MODEL FAMILY	STATUS	ITEM
Cap-tier classifiers (large, mid, small)	Built	A5
Setup models: pullback, breakout, momentum, reversal	Planned	F07
Horizon and stop-first heads	Planned	F05
Meta decision model (calibrated)	Planned	F04
Exit model	Planned	F13
Regime detection	Rule table (by design)	A1
Anomaly and shock detection	Planned	F09

Strategies are already plug-ins: STRATEGY_REGISTRY with @register_strategy (4 today: SMA crossover, ML swing, long-term value, tier). Strategy configuration lives in strategies.params; risk fields can't be set per strategy.

20 Portfolio intelligence and execution

Portfolio intelligence

- Today: sector cap, cap-tier budget, cash floor, deployable ceiling, capital ladder
- Planned: health score, correlation clusters, factor exposure, opportunity cost, optimisation
- Portfolio risk can overrule a strong single signal

Execution modes

Paper	Built (default)
Shadow (advisory)	Built
Assisted live (approve each order)	Planned
Auto live	Latch built; blocked by after-close order method and GTT stops

Order states tracked: submitted, accepted, rejected, partly filled, filled, cancelled, reconciled (reconcile_orders every 60 s; holdings_sync daily). Execution stays separate from decisions: a broker adapter turns an order intent into Kite's API.

21 Screens

SCREEN (BLUEPRINT)	TODAY'S PAGE	PLANS	ITEMS
Home / Dashboard	Dashboard.tsx	All	F21, F01
Today's signals / Scanner	ScanResults.tsx	All, shaped	F10, F44
Signal detail / Stock page	StockDetailModal.tsx, PricePerformance.tsx	All, shaped	F47
Position tracker	Positions.tsx, Holdings.tsx	Plus, Pro	F12, F50
Portfolio and capital allocation	Capital.tsx	Plus, Pro	F15
Market regime and shock centre	—	All, shaped	F01, F36, F09
Strategy explorer	Strategies.tsx	All, shaped	F27
Model and strategy health	ModelLab.tsx, Models.tsx	Plus, Pro	F23
Reports	Reports.tsx, RealReport.tsx	All, shaped	F37
Experience / why this happened	—	Pro	F35, F31
Decision replay	—	Owner, Pro	C2
AI copilot	—	Pro	F20
Alerts and broker connections	Settings.tsx (Kite login)	All	P5, P1
Settings / risk policy, safety	Settings.tsx, Safety.tsx	All / owner	P6
Admin, plans, audit	Plans page (branch)	Owner	P6
Upgrade and sign-up	AuthScreen.tsx	All	F48, F49
Personal finance and mutual funds	Finance.tsx	Owner (frozen)	X1, X2

22 AI copilot, benchmarks and attribution

The copilot is an explanation and navigation layer over the same contracts, never a separate brain. It answers from decision records, evidence and timestamps through read-only tools, cites its sources, has a daily cap, and gives no personal order instructions.

QUESTION IT ANSWERS	READS
Why this stock? Why WAIT and not TRADE?	Decision record, evidence
What changed today? Why did confidence fall?	What-changed records
Why are we defensive?	Market state, regime events
What happened in similar situations?	Market and strategy experience
Why was an exit triggered?	Exit record, outcome
What is my portfolio exposure?	Portfolio health, correlation

Benchmarks: NIFTY, sector indices and the SMA-crossover baseline (already in the code as the benchmark strategy). **Attribution** separates market, sector, stock selection, timing, execution and costs when data allows, so we know whether the system adds value or just rides a rising market.

23 Reliability, observability and security

FAILURE	SAFE BEHAVIOUR	TODAY
Cache down	Fall back to PostgreSQL	Redis is not truth
Data provider down	Mark data unavailable; decisions WAIT	Alerts; quality score planned
Model failure	No signals (HOLD) or safe fallback	No active model → no decision
Broker disconnect	No false execution assumptions; re-login and retry	Re-login and retry built
Queue failure	Retry and dead-letter list	Planned
Server outage	Broker-held stops protect positions	GTT planned

Observability metrics

Data freshness, ingestion lag, scan duration, model latency, API latency, database health, broker connectivity, order latency, signal volume, model drift, strategy performance, expectation drift, decision disagreement, incidents. Every event carries `decision_id`, `trade_id` and `run_id`.

Security and governance

Roles (owner, admin, support, user), least privilege, secrets outside git (`/root/app/.env`), encrypted broker tokens, audit logs, environment separation (local, staging, production), explicit paper / live modes, emergency stop, consent records, data-protection review.

24 Independence and repository structure

Ports and adapters, grown inside today's repository as a modular monolith. No big-bang move: new code goes into the new shape, old code moves when touched.

TARGET FOLDER	PURPOSE	TODAY'S CODE THAT BELONGS THERE
domain/	State, situation, decision, risk, portfolio, experience rules	services/risk.py, limits.py, exit_policy.py, deployable.py, avoid.py, core/strategy_policy.py
intelligence/	Models, features, strategies, reasoning	ml/, strategies/, services/factors.py
application/	Scans, tracking, re-evaluation, learning	services/engine.py, execution.py, backtest.py, portfolio.py, real_report.py
ports/	Interfaces: market data, repository, cache, broker, notifications, model	brokers/base.py (Broker port, already here in spirit)
adapters/	Database, cache, broker, external providers	brokers/kite.py, paper.py, notifications/telegram.py, market_feeds.py, ingestion.py
api/, workers/	REST and scheduled jobs	Already exist
observability/	Logging, metrics, tracing	core/logging.py, workers/heartbeat.py
config/	Versioned policies and flags	core/config.py, core/plans.py (branch)
brain/ (new first)	Contracts, registry, pipeline, decision	Wraps the above behind contracts

Vendor-specific today: candles use a TimescaleDB hypertable, so a database change is a planned migration, not a switch. Broker, cloud (Docker) and screens (API-driven React) are already replaceable.

25 Data model

GROUP	TABLES TODAY (34)	NEW TABLES PLANNED
Market	instruments, candles, quotes	market_state_daily, regime_events, sector_rank_daily, theme_rank_daily
Feeds	daily_delivery, block_deals, institutional_flows, upcoming_events, trading_restrictions	event_items, event_impacts, shock_events
Trading	strategies, signals, orders, positions, trades, portfolio_snapshots, broker_fills	decision_records, setup_detections, position_thesis, trade_journal; signals.is_shadow
ML	ml_models, predictions	model_evaluations, experiments, feature_store tables
Memory	—	market_episodes, episode_replays, stock_profile_daily, score_band_stats, filter_scorecard_daily, lessons_reports, incidents
Safety	system_state, risk_events	constitution_versions, approvals
Users and plans	users, broker_sessions, subscription_plans, plan_settings (branch)	portfolios, watchlists, broker_connections, subscriptions, coupons, invoices, consents, audit_log, admin_actions, invites
Personal finance (frozen)	finance_ingested_files, finance_transactions, finance_loans, finance_recurring_bills, finance_recurring_bill_payments, finance_daily_categories, finance_custom_rules, mutual_funds, mutual_fund_navs, mutual_fund_holdings	—

26 API surface

AREA	TODAY (API/V1/ENDPOINTS)	PLANNED
Auth and users	auth	Sign-up, OTP (go-live), /me/*
Market and data	market_data, instruments	/market/today, /market/memory, /market/sectors, /market/themes
Signals and decisions	signals, strategies	/stocks/{symbol}/page, /stocks/{symbol}/decision, /changes/today
Trading	orders, portfolio, risk, safety, real_report	/me/positions/intel, /me/portfolio/health
ML and research	ml, backtest	/scorecards/*, /research/runs, /ml/models/{id}/scorecard
System	system, notifications	/system/data-quality, incidents
Plans and admin	plans (branch)	/admin/*, /billing/*
Personal finance (frozen)	finance, mutual_funds	—

27 Cloud sizing and scaling tiers

Sizing must come from real load tests. Today everything runs on one droplet (five containers). The tiers below map the blueprint's baseline onto our months.

TIER	WHEN	SHAPE
A · Development	Always	Docker Compose locally (PostgreSQL on 5433), no live trading
B · MVP and paper	Months 1–4	Today's droplet + object storage backups + a staging environment
C · Production, assisted	Month 4 onwards	Trading core (droplet, static IP) + product stack: API/web 2–4 vCPU, 4–8 GB; worker 4–8 vCPU, 8–16 GB; managed PostgreSQL 2–4 vCPU, 8–16 GB SSD; Redis 1–2 vCPU, 2–4 GB; object storage; monitoring; on-demand training
D · Scale	Only when load demands	More worker pools, read replicas, autoscaling, dedicated ML serving, disaster recovery

28 Testing and release lifecycle

TEST TYPE	TODAY	ADD
Unit tests for domain rules	60+ test files (risk, limits, exits, features, calibration...)	Contracts, decision states
Integration (DB, broker)	PostgreSQL + TimescaleDB in CI	Broker contract tests
Walk-forward and backtest	Purged walk-forward, backtest tests	Strict availability times; bias checks
Replay of historical decisions	Research scripts	Decision replay tests; byte-identical scan snapshot
Failure injection	—	Cache, DB, feed, broker, model-file failures
UI and API	Typecheck and build in CI	Plan-gate and isolation tests
Migrations	Alembic on startup	Rollback rehearsal

Release lifecycle

Design → implement → unit tests → integration tests → historical replay → backtest
→ paper → shadow → limited rollout → monitor → promote or roll back

A new model or strategy never silently replaces a stable one.

29 Everything the application has today

SCHEDULED JOBS (WORKER)	TIME	AREA	WHAT EXISTS
heartbeat, check_kite_login, refresh_quotes, check_exits, reconcile_orders	every 60 s	Screens (13)	Dashboard, Scan results, Strategies, Models, Model Lab, Positions, Holdings, Capital, Safety, Reports, Real report, Settings, Finance (+ Plans on the branch)
kite_auto_login	06:10	API files (17)	auth, backtest, finance, instruments, market_data, ml, mutual_funds, notifications, orders, portfolio, real_report, risk, safety, signals, strategies, system (+ plans)
nag_missing_session	06:15–15:45	CLI (13)	init-db, create-user, make-owner, kite-login, sync-instruments, sync-index, sync-context-indices, backfill, train, scan, backtest, status
daily_market_feeds	08:30, 19:20, 22:05	Strategies (4)	SMA crossover (benchmark), ML swing, long-term value (advisory), tier
daily_ingest	15:40	Brokers	PaperBroker, KiteBroker behind one interface; two-flag live latch
predict_watchlist	15:42	Apps	Web, installable PWA, Android and iOS wrappers
signal_scan	15:45	Personal finance	Bank and PhonePe statement parsers, categoriser, loans, recurring bills, CAS mutual-fund import, NAV sync (frozen)
sync_real_holdings	15:48	Operations	CI tests, SSH deploy, rollback images, database backup on deploy, heartbeat verification, model artifact backup
capture_fills	15:50, 19:30		
daily_summary	16:00		
evaluate_predictions, evaluate_signals	16:15, 16:17		
sync_mutual_fund_navs	21:30		
sync_instruments	Sunday 08:00		

30 Complete requirements register

Every item: what exists (A), what extends existing code (B), what's new (C), what's future (D), plus the existing modules. The last column maps to the blueprint's REQ numbers.

ID	ITEM	STATUS	PRIORITY	WHEN	FREE / PLUS / PRO	BLUEPRINT
Market						
A1	Regime and deployable capital	A	Built	Built	✓ / ✓ / ✓	REQ-003
A4	NSE information feeds	A	Built	Built	internal	REQ-001
F01	Market intelligence layer	B	Core	Month 1	✓ / ✓ / ✓	REQ-002
F02	Sector brain	B	Core	Month 2	— / ✓ / ✓	REQ-002
F26	Regime change detector	B	Core	Month 1	— / ✓ / ✓	REQ-003, REQ-012
F35	Market memory (episode library)	C	Core	Month 1	✓ / ✓ / ✓	REQ-004, REQ-005
F36	Bear mode and playbook	C	Core	Month 2	✓ / ✓ / ✓	
F42	Big-investor deals	B	Diff	Month 3	✓ / ✓ / ✓	
F46	Theme rotation	C	Diff	Month 3	✓ / ✓ / ✓	REQ-002
C8	Market state machine: NORMAL, CAUTION, DEFENSIVE, SHOCK	B	Core	Month 1	✓ / ✓ / ✓	REQ-002, REQ-003
F59	Insider, promoter and pledge disclosures (new)	B	Diff	Month 3	— / ✓ / ✓	
Portfolio						
A2	Risk engine and capital ladder	A	Built	Built	— / ✓ / ✓	REQ-007, REQ-008, REQ-009
A6	Exit policy	A	Built	Built	✓ / ✓ / ✓	REQ-013
F12	Position intelligence	B	Diff	Month 4	— / ✓ / ✓	REQ-011, REQ-012
F13	Exit intelligence	B	Diff	Month 5	✓ / ✓ / ✓	REQ-013
F14	Opportunity cost engine	C	Diff	Month 6	— / — / ✓	
F15	Portfolio health	B	Diff	Month 4	— / ✓ / ✓	REQ-009
F16	Correlation engine	C	Core	Month 2	— / — / ✓	
F17	Factor exposure	B	Diff	Month 4	— / — / ✓	
F30	What-if simulator	C	Moat	Months 7–12	— / — / ✓	
F33	Advanced portfolio optimisation	D	Moat	Months 7–12	— / — / ✓	REQ-009, REQ-105
F41	Structure stops, targets and time stop	C	Core	Month 2	— / ✓ / ✓	

ID	ITEM	STATUS	PRIORITY	WHEN	FREE / PLUS / PRO	BUEPRINT
F43	Community diversification benchmark	C	Diff	Month 6	— / ✓ / ✓	
F45	Tax view	C	Diff	Month 4	✓ / ✓ / ✓	
Decision						
A3	Landmine filter	A	Built	Built	✓ / ✓ / ✓	
F04	Meta confidence system	B	Core	Month 3	✓ / ✓ / ✓	
F06	Expected value engine	C	Core	Month 2	— / ✓ / ✓	
F10	“Why not buy?” engine	B	Diff	Month 1	✓ / ✓ / ✓	
F11	“What changed?” engine	B	Diff	Month 3	— / ✓ / ✓	REQ-012
F22	Explainability	B	Diff	Month 3	✓ / ✓ / ✓	
F24	Data quality	B	Core	Month 1	✓ / ✓ / ✓	REQ-001, REQ-020
F25	Uncertainty and model agreement	C	Core	Month 3	✓ / ✓ / ✓	
F40	Trigger entry	C	Core	Month 1	— / ✓ / ✓	
Learning						
A5	Calibration, walk-forward, Model Lab	A	Built	Built	— / ✓ / ✓	REQ-017
A7	Backtest replay	A	Built	Built	internal	
F18	Trading journal	B	Core	Month 1	✓ / ✓ / ✓	REQ-005
F23	Model observatory	B	Diff	Month 6	— / ✓ / ✓	REQ-017, REQ-113
F27	Strategy lab	B	Diff	Month 6	✓ / ✓ / ✓	REQ-014, REQ-018
F28	Research lab	B	Diff	Month 6	— / — / ✓	REQ-015, REQ-104
F29	Monte Carlo stress testing	C	Moat	Months 7–12	— / — / ✓	
F31	Failure investigator	C	Core	Month 5	— / — / ✓	REQ-016
F32	Automated challenger loop	D	Moat	Months 7–12	internal	REQ-113
F37	Score report card	B	Core	Month 2	✓ / ✓ / ✓	REQ-014
F38	Filter and factor scorecard	B	Core	Month 2	— / — / ✓	REQ-104
F39	Lessons loop	C	Core	Month 3	— / ✓ / ✓	
Platform						
A8	Broker, alerts, real-account tracking	A	Built	Built	internal	REQ-019, REQ-110

ID	ITEM	STATUS	PRIORITY	WHEN	FREE / PLUS / PRO	BLUEPRINT
F48	Upgrade and checkout page	C	Platform	Month 4	internal	
F49	Sign-up and sign-in pop-up	B	Platform	Month 3	internal	
P1	Accounts, portfolios, broker connections	C	Platform	Month 3	internal	
P2	Subscriptions and entitlements	C	Platform	Month 4	internal	
P3	Compliance module	C	Platform	Month 4	internal	REQ-020
P4	Licensed market data	C	Platform	Month 5	internal	
P6	Admin panel	B	Platform	Month 4	internal	
P7	Deployment split and staging	C	Platform	Month 4	internal	REQ-111
P5	Notification hub	B	Platform	Month 4	internal	
C9	Broker-held stops (GTT) and reconciliation v2	B	Platform	Month 4	internal	REQ-019
C12	Observability metrics and trace IDs	B	Platform	Month 1	internal	REQ-020
C14	Review and merge feature/plans-manager	B	Platform	Month 3	internal	
F58	Order hand-off to Kite basket (new)	C	Platform	Month 4	— / ✓ / ✓	
F60	Public stock pages for search engines (new)	D	Platform	Months 7–12	✓ / ✓ / ✓	
C16	Execution quality: slippage and fills (new)	B	Platform	Month 4	internal	
Stock						
F03	Stock DNA	C	Core	Month 4	— / ✓ / ✓	
F05	Time-horizon probabilities	B	Core	Month 4	— / — / ✓	REQ-010
F07	Setup-specific models	C	Core	Month 2	— / ✓ / ✓	
F08	News classification	C	Core	Month 5	✓ / ✓ / ✓	REQ-103
F09	Event shock detector	C	Diff	Month 5	— / — / ✓	
F34	Event-impact learning	D	Moat	Months 7–12	— / — / ✓	REQ-103
F44	Ready-made screens	C	Diff	Month 4	✓ / ✓ / ✓	
F47	Stock page	B	Diff	Month 1	✓ / ✓ / ✓	REQ-108
F56	Signal explorer with setup filters and risk label (new)	B	Core	Month 2	✓ / ✓ / ✓	
F57	Similar past setups table (new)	C	Diff	Month 2	✓ / ✓ / ✓	
User						

ID	ITEM	STATUS	PRIORITY	WHEN	FREE / PLUS / PRO	BLUEPRINT
F19	Personal trading twin	D	Moat	Months 7–12	— / — / ✓	
F20	AI copilot	D	Diff	Month 6	— / — / ✓	REQ-109
F21	Morning brief	C	Diff	Month 2	✓ / ✓ / ✓	
F52	After-investing screen and checklist	C	Diff	Month 4	✓ / ✓ / ✓	
F53	Onboarding wizard (new, 29 Sep designs)	C	Platform	Month 3	✓ / ✓ / ✓	
F54	Personal preferences and tighten-only risk presets (new)	C	Diff	Month 3	✓ / ✓ / ✓	
F55	Watchlist (new)	C	Core	Month 2	✓ / ✓ / ✓	
Positions						
F50	Expectation envelope and live trade tracking	C	Diff	Month 4	✓ / ✓ / ✓	REQ-010, REQ-011
F51	Exit scenarios card	C	Diff	Month 5	— / ✓ / ✓	REQ-013
Brain platform						
C1	Nine stable brain contracts	C	Core	Month 1	internal	REQ-001, REQ-101
C2	Decision contract, ledger and replay	B	Core	Month 1	internal	REQ-006, REQ-015
C3	Time truth: five timestamps	B	Core	Month 1	internal	REQ-001
C4	Feature store (dated tables)	C	Core	Month 2	internal	
C5	Bias checks: look-ahead, survivorship, overfitting	C	Core	Month 2	internal	
C6	System incident memory	C	Platform	Month 3	internal	REQ-016, REQ-112
C7	Versioned constitution module	B	Core	Month 1	internal	REQ-007
C11	Ports and adapters, grown in place	B	Platform	Month 1	internal	
C13	Experience memory store (eight memories)	C	Core	Month 2	internal	REQ-005
C15	AI (LLM) use policy (new)	C	Core	Month 5	internal	
Existing modules						
X1	Personal finance module (statements, loans, bills)	A	Built	Built	internal	
X2	Mutual fund tracking (CAS import, NAVs)	A	Built	Built	internal	
X3	Long-term stock picks (advisory strategy)	A	Built	Built	internal	

Blueprint requirement → our items

BLOCK	BLUEPRINT	REQUIREMENT	OUR ITEMS
	REQ-001	Input contracts and provenance	C1, A4, F24, C3
	REQ-002	State engine	F01, F02, F46, C8
	REQ-003	Market-regime detection	A1, F26, C8
	REQ-004	Situation recognition	F35
	REQ-005	Experience memory	F18, F35, C13
	REQ-006	Decision contract	C2
	REQ-007	Constitution	C7, A2
	REQ-008	Risk engine	A2
	REQ-009	Capital allocation	F15, F33, A2
	REQ-010	Expectation engine	F50, F05
	REQ-011	Position tracking	F12, F50
	REQ-012	Re-evaluation	F11, F12, F26
	REQ-013	Exit intelligence	A6, F13, F51
	REQ-014	Benchmark attribution	F27, F37
	REQ-015	Decision replay	C2, F28
	REQ-016	Failure intelligence	F31, C6
	REQ-017	Model registry	A5, F23
	REQ-018	Strategy registry	F27
	REQ-019	Broker reconciliation	A8, C9
	REQ-020	Observability and audit	F24, P3, C12
	REQ-101	New data-source plug-in	C1
	REQ-102	Alternative data	Excluded: cost, licensing, little value for swings
	REQ-103	Advanced news intelligence	F08, F34
	REQ-104	Counterfactual analysis	F28, F38
	REQ-105	Advanced portfolio optimiser	F33
	REQ-106	Institutional multi-user governance	Later, after legal review
	REQ-107	API / white-label access	Later, after legal review
	REQ-108	Mobile interface	F47 (PWA and app wrappers exist)
	REQ-109	Advanced AI copilot	F20
	REQ-110	Multi-broker execution	A8
	REQ-111	Multi-cloud deployment	P7
	REQ-112	Automated incident learning	C6
	REQ-113	Advanced champion / challenger	F23, F32

31 Implementation plan

The blueprint's phases 0–4, aligned with our months, with the three gates kept. Tests, replay, monitoring, rollback and documentation are part of every feature.

PHASE	WHEN	BUILD	GATE
0 · Foundations	Month 1, weeks 1–2	Brain contracts (C1); decision contract and ledger (C2); time truth (C3); constitution module (C7); observability IDs (C12); wrap today's engine as plug-ins	Scan output identical to today with flags off
1 · Data, state, signals	Months 1–3	Market brain and state machine (F01, C8); data quality (F24); why / why not (F10); journal and shadow scoring (F18, F38); setup detectors and models (F07); Expected R (F06); trigger entry test (F40); stock page (F47); accounts and plans branch merge (P1, C14)	Edge answer; setups and EV beat the baseline on paper
2 · Expectation and experience	Months 2–5	Expectation envelope and tracking (F50); re-evaluation (F11, F12); market memory and bear mode (F35, F36); failure intelligence (F31, C6); decision replay (C2); scorecards (F37, F38); lessons (F39)	Calibration gap $\leq$ 10 points; per-episode loss report
3 · Portfolio, defence, platform	Months 4–6	Portfolio brain (F15–F17); opportunity cost (F14); exit model (F13); champion / challenger (F23); billing, admin, sign-up (P2, P6, F48, F49); stack split and staging (P7); licensed data (P4); GTT stops (C9)	Free can't read Pro data; launch decision
4 · Advanced	Months 7–12	Copilot (F20); research lab, Monte Carlo, what-if (F28–F30); twin (F19); event-impact learning (F34); optimisation (F33); multi-broker and fundamentals	Per-feature gates

Month 1, week by week

WEEK	CODE	RESEARCH AND OWNER ACTIONS
1	brain/contracts.py (nine contracts); decision_records migration with time truth; adapter from SignalDecision to Decision	Consultant engaged; data-vendor quotes requested
2	Wrap the engine behind brain/pipeline.py; byte-identical snapshot test; constitution module reading today's locked rules	Edge investigation replays: scale-out off, signal exit off, higher thresholds, next-open and trigger entry
3	Shadow scoring (signals.is_shadow); journal rows; data-quality score; market state and regime events	Episode detection; replay of the 2021–22 and 2024–25 corrections
4	Stock page v1 (header, decision, chart levels, why / why not) reading decision records	Month 1 gate review; NISM exam booked

32 Stable, never hard-coded, and flexibility

NEVER HARD-CODE	TODAY
Broker calls in decision logic	Avoided (Broker port)
Database queries in domain logic	Mixed; move behind repositories
Redis as truth	Avoided
One fixed confidence formula	One threshold today; meta-model planned
One fixed regime list	Rule table in one module
One fixed strategy	Registry
One cloud provider	Docker, portable
"Every market is tradeable"	Deployable %, drawdown halt

CHANGE	HOW IT'S ABSORBED
New screen or app	No brain change
New database	Adapter + planned migration
New cache	Adapter
New broker	One Broker class
New data provider	Input adapter + contract
New model	Model contract + registry
New strategy	@register_strategy plug-in
New decision type	Versioned decision contract
New brain capability	Module through stable contracts

33 Review checklist, answered for today

QUESTION	ANSWER TODAY	TO CLOSE IT
Are all inputs timestamped and validated?	Partly	Time truth (C3), gateway checks
Can every decision be replayed?	Partly	Decision ledger with versions (C2)
Can the cache disappear without corrupting state?	Yes	—
Can the database be migrated?	Yes (Alembic)	Rollback rehearsal
Can a model be rolled back?	Yes (registry)	—
Can a broker mismatch be detected?	Partly	Reconciliation v2, incidents
Can TradeMind say UNKNOWN / NO-TRADE?	Partly	Decision states (chapter 12)
Are expected vs actual outcomes tracked?	Partly	Expectation engine (F50)
Are benchmark comparisons available?	Partly	NIFTY and sector attribution
Are safety rules versioned?	Locked, not versioned	Constitution versions (C7)
Can the UI be replaced?	Yes	—
Can capabilities be added without changing contracts?	Strategies yes	Brain contracts (C1)
Are failures remembered?	Alerts only	Incident memory (C6)

34 Top risks

RISK	SEVERITY	MITIGATION
No proven profit after costs	High	Edge investigation first; launch gate
SEBI registration delay; personal advice rules	High	Start Month 1; legal opinion; reviews not instructions
Expected returns read as promises	High	Probabilities with sample size; "plan this trade"
Look-ahead and survivorship bias	High	Time truth, bias checks, dated universe
Overfitting from many models	High	Holdout, walk-forward, paper first
Scope for one developer with AI help	High	Contracts first; one plug-in at a time
Outage leaves positions without stops	High	GTT stops; product stack split
Data licences and costs	Medium	Quotes in Month 1; Free on end-of-day data
Over-engineered infrastructure	Medium	Small set; add on measured need
Language model invents facts	Medium	Citations; evaluation sets

35 Problem log and decisions

ID	PROBLEM OR QUESTION	AREA	PRIORITY	RECOMMENDED NEXT ACTION	STATUS
P-001	Decision vocabulary: adopt the three sets with UNKNOWN (chapter 12)?	Brain	High	Approve	Open
P-002	Holding period: up to 15 trading days, or shorter?	Strategy	High	Keep 15 (model's label); show odds by horizon	Open
P-003	Keep locked stop -4% / target +8%, or setup-specific levels?	Risk	High	Keep until replay proves otherwise (F41)	Open
P-004	No proven profit after costs in replay	Edge	High	Month 1 edge investigation	Open
P-005	SEBI Research Analyst registration	Compliance	High	Consultant, NISM exam, application	Open
P-006	Personalised recommendations and Investment Adviser rules	Compliance	High	Legal opinion	Open
P-007	Licensed market data for users	Data	High	Vendor quotes in Month 1	Open
P-008	Broker execution for users	Execution	Medium	Out of year one	Open
P-009	Merge feature/plans-manager	Platform	Medium	Review in Month 3 (or earlier)	Open
P-010	How to show expected returns	UX / compliance	High	Probabilities with sample size only	Decided
P-011	Market states vs today's six regimes	Brain	Medium	Map as in chapter 16	Open
P-012	Infrastructure scope (Kafka, Kubernetes, extra databases)	Infrastructure	Medium	Small set; add on need	Open
P-013	Point-in-time fundamentals source	Data	Medium	NSE XBRL in the 12-month plan	Open
P-014	Broker-held stops (GTT)	Reliability	High	Build in Month 4	Open
P-015	After-close entry method (AMO or next-open trigger)	Execution	High	Test trigger entry (F40) in Month 1	Open
P-016	Survivorship bias in the replay universe	Research	Medium	Universe by date; bias checks (C5)	Open
P-017	Daily Kite login for users' read-only links	Integration	Low	Confirm with Zerodha	Open
P-018	Language model provider and cost caps	AI	Medium	Claude API; daily cap per Pro user	Open
P-019	White-label, B2B, institutional use	Business	Low	Later, after legal review	Open
P-020	Personal finance and mutual fund modules in the product?	Scope	Low	Keep frozen and owner-only	Open

ID	PROBLEM OR QUESTION	AREA	PRIORITY	RECOMMENDED NEXT ACTION	STATUS
P-021	Design mock-ups use -0.8% / -1.9% stops, +4% targets and 5 days; the engine uses -4%, +8% and up to 15 days	Design	High	Keep engine levels in designs until F41 replay decides	Open
P-022	Users placing orders from inside TradeMind	Execution / compliance	High	Kite basket hand-off (F58); confirm with Zerodha and the consultant	Open
P-023	Performance numbers in designs (62% win rate, profit factor 2.4, +18.6%)	Compliance	High	Verified numbers only; label mock-ups as illustrative	Decided
P-024	AWS reference deployment vs DigitalOcean	Infrastructure	Medium	Stay on DigitalOcean until Tier D	Open
P-025	Microsoft sign-in shown on the login screen	Platform	Low	Not now; Okta covers it at go-live	Open
P-026	AI detecting unknown situations and generating strategies	Brain	Medium	Statistical novelty detection; AI ideas are proposals only	Open
P-027	Next.js in the architecture vs today's React/Vite app	Frontend	Low	Keep Vite; public SEO pages later (F60)	Open

A planning document. This blueprint is not a promise of trading performance or returns. Expected-return ranges and probabilities must be statistically validated before they are shown as decision-grade numbers. Illustrative values are examples, not model output.

36 Review: your Central Brain picture

Figure 36.1 · TradeMind Brain: Central Intelligence Brain your picture, 29 Sep



Verdict: the clearest brain picture so far. It fits the design in Parts B and C. Keep the eight-step pipeline, the six modules with their guiding questions, the plug-in row, “alternative decisions”, “decision expiry”, “scenario analysis”, “benchmark memory” and the two panels showing where ML and AI are used. Eleven points need a change before it becomes the reference picture.

#	AREA	YOUR PICTURE SHOWS	CHANGE	WHY
1	Decision words	BUY / SELL / HOLD / WAIT in the engine; TRADE / WATCH / WAIT / AVOID in outputs; Buy / Sell / Hold / Rebalance for portfolios	Use the three sets from chapter 12, including UNKNOWN	One vocabulary on every screen; no short selling, so no SELL signal
2	Where safety sits	Risk and safety is module 5, before the decision	Keep module 5 for early checks, and let the constitution check the final decision again, last	Sizing and user settings come after module 5 and could otherwise break a locked limit
3	Regime classification	An ML model	The rule table stays the regime of record; an ML regime model runs only in shadow	Rules are built, auditable and don't need many past regimes to train

#	AREA	YOUR PICTURE SHOWS	CHANGE 	WHY
4	Detecting unknown situations	Listed as an AI (LLM) job	Measure it: distance to the nearest past episode plus model disagreement. The LLM only explains the result	Language models can't reliably measure how new a situation is
5	Generating strategies	Listed as an AI (LLM) job	LLM ideas become research proposals, then backtest, shadow run and owner approval	Principle 2: no uncontrolled changes
6	Learning loop	"Adjust confidence", "adjust risk parameters", "adapt to new regimes"	Proposals only. Risk parameters never change automatically; recalibration goes through shadow and approval	Principle 1: safety rules never change on their own
7	Real-time feature store	Features computed live	Daily feature tables stored by date (C4); intraday prices only for exits and alerts	Swing decisions are made after the close; live features add cost and look-ahead risk
8	Inputs	Futures, options, order book; social sentiment, satellite data, insider activity, options flow	Out for now: futures, options, order book, social, satellite, options flow. In: insider and promoter disclosures (new F59)	Licence cost against little value for delivery swings; insider data is free from NSE
9	Trade execution output	"Place orders via broker (Zerodha/others)"	Only for your own book. Users hand the order to Kite and confirm it there (new F58)	SEBI adviser and algo-trading rules; users stay in control
10	User behaviour memory	Overrides and preferences	Used to personalise what a user sees; never used to train the models	Learn from outcomes, not clicks (principle 10)
11	Missing from the picture	—	Keep from Figure 3.2: the constitution as an outer layer, time truth, decision ledger, incident memory, system health and Free / Plus / Pro	They're in the design; the picture should show them

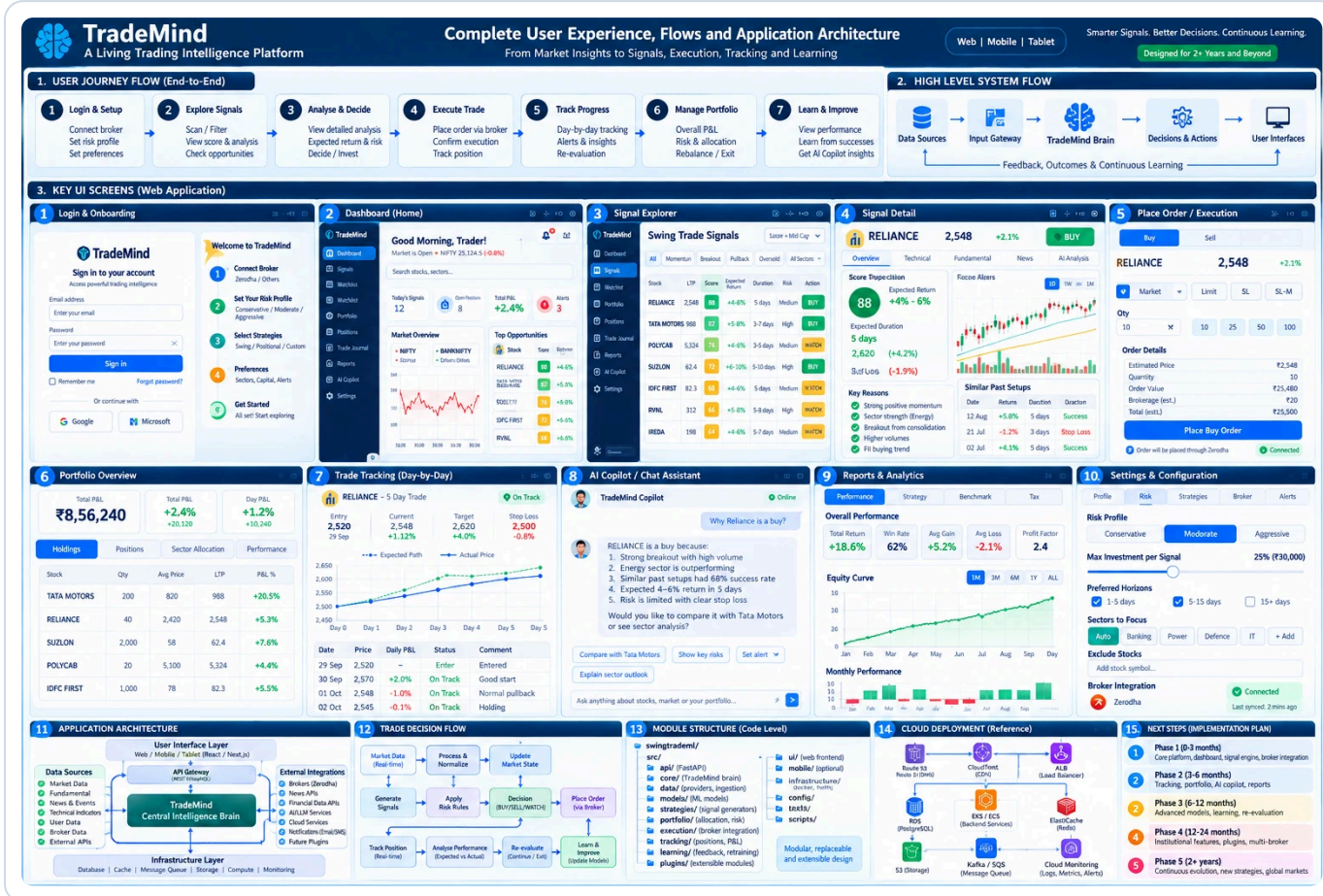
Where ML and AI are used: your two panels checked

USE IN YOUR PICTURE	OUR POSITION	ITEM
Machine learning		
Market regime classification	Rules of record; ML in shadow	A1, F26
Price movement prediction	Built: three cap-tier models	A5, F05, F07
Feature importance	Model Lab; scorecards planned	A5, F38
Pattern recognition	Setup detectors: rules first, then models	F07
Anomaly detection	Planned	F09

USE IN YOUR PICTURE	OUR POSITION	ITEM
Risk estimation (volatility, drawdown, stop probability)	Planned	F05, F15
Portfolio optimisation	Months 7–12	F33
Model ensemble	Planned: meta-model and agreement	F04, F25
AI (large language model)		
News and event analysis	OK, with rules first and citations	F08
Company report analysis	OK, with citations	F08, F03
Explainability	OK, only from the decision record	F22
AI copilot	OK, read-only tools and a daily cap	F20
Sentiment analysis	News only; social media out in year one	F08
Summarisation	OK	F21
Detect unknown situations	Change: measured statistically	F35, F25
Generate strategies	Change: proposals only	F28

37 Review: your Experience and Application picture

Figure 37.1 · Complete User Experience, Flows and Application Architecture your picture, 29 Sep



Verdict: the ten screens are a strong base for the product and match most of the requirements. The main problems are the numbers in the mock-ups: they contradict today's engine and each other. Also, users placing orders from inside TradeMind isn't possible in year one.

SCREEN	MATCHES	CHANGE NEEDED 
1 · Login and onboarding	F49, new F53	Drop “Microsoft” for now (password and Google now; phone OTP and Okta at go-live). The broker link is read-only, and only for Plus and Pro. The risk profile can only make limits stricter
2 · Dashboard	F21, F01	Add the market-state banner (NORMAL / CAUTION / DEFENSIVE / SHOCK). The signal count follows the plan (Free sees 5)
3 · Signal explorer	F44, new F56	The action column uses TRADE / WATCH / WAIT / UNKNOWN / AVOID. Show Expected R and odds instead of “+4–6%”. Duration is “up to 15 days”. Define the risk label from volatility and stop distance
4 · Signal detail	F47, new F57	Levels come from the engine. Show the score with its band's real hit rate and sample size. “BUY” becomes “Plan this trade”. Similar past setups include the losers and show how many there are
5 · Place order	New F58	For users: an “Open in Kite” basket they confirm at Zerodha. TradeMind's own order form is only for your book. No short sells
6 · Portfolio overview	F15	Add the health score and sector-cap warnings

SCREEN	MATCHES	CHANGE NEEDED ↘
7 · Trade tracking	F50	Show the expected path as a band (typical range), not a single line. Stop and target must match the signal. Use the four drift states
8 · AI copilot	F20	"RELIANCE is a buy... expected 4–6% in 5 days" becomes "scores TRADE because...; of 120 similar setups, 58% reached +5% first; not a promise". Cite the decision record
9 · Reports	F37, F45	Show only verified numbers. The mock-up's 62% win rate, 2.4 profit factor and +18.6% are not ours: today's replay shows a 40–45% win rate and flat returns. Compare against NIFTY
10 · Settings	New F54, P5	"Max per signal" can only be lower than the ladder cap. The 1% risk stays locked. The 1–5 day horizon needs F05; 15+ days is the long-term strategy

The mock-up numbers against the engine

FIELD	SIGNAL DETAIL (SCREEN 4)	TRADE TRACKING (SCREEN 7)	ENGINE TODAY
Stop	−1.9%	₹2,500 on ₹2,520 = −0.8%	−4% from entry
Target	₹2,620 (+4.2%)	₹2,620 (+4.0%)	+8%, with half booked at +5%
Holding period	5 days	5 days	Up to 15 trading days; 30-day time stop
Entry	₹2,548	₹2,520	Trigger price on the next session

A −0.8% stop sits inside a normal day's move for most stocks, so it would be hit by ordinary noise. Keep the engine's levels in every design until the structure-stop replay (F41) proves better levels. That's problem P-021.

Sections 11–15 of your picture

SECTION	KEEP	CHANGE ↘
11 · Application architecture	Layers, external integrations, plug-ins	Keep today's React/Vite app, not Next.js (public pages for search engines later, F60). The API gateway is FastAPI behind a reverse proxy. Use a Redis queue, not a separate message broker
12 · Trade decision flow	Market data → state → signals → risk → decision → track → re-evaluate → learn	Use chapter 12's words. The safety check runs last. The broker order is only for your book
13 · Module structure	The idea of one folder per module	Map it onto today's repo (chapter 24) rather than renaming. <code>models/</code> already means database tables; ML lives in <code>mL/</code> . Terraform is optional later
14 · Cloud deployment	A useful reference	Stay on DigitalOcean until Tier D (table below). An AWS stack with Kubernetes and Kafka costs several times more and needs operations work we don't need yet
15 · Next steps	Phases 4 (12–24 months) and 5 (2+ years) as the long-term view	Phases 1–3 follow chapter 31, which adds the edge investigation, SEBI and contracts first. Broker order placement for users moves out of Phase 1

AWS IN YOUR PICTURE	WHAT WE USE INSTEAD	WHEN
Route 53 · CloudFront	DNS and CDN through Cloudflare or DigitalOcean	Launch
Application load balancer	DigitalOcean load balancer	When there are 2+ API servers
EKS / ECS	Docker Compose on droplets (as today)	Kubernetes only at Tier D
RDS PostgreSQL	Self-hosted PostgreSQL + TimescaleDB with backups, or managed PostgreSQL once TimescaleDB support is confirmed	Month 4 decision
ElastiCache Redis	Redis container, or managed Redis later	As today
S3	DigitalOcean Spaces (works like S3)	Month 3, for backups and models
Kafka / SQS	Redis queue with retries and a dead-letter list	Month 3
Cloud monitoring	Uptime checks, heartbeat, logs and metrics	Month 1 onwards
—	Reserved static IP for Kite (the trading core has it already)	Kept

38 What changes in the requirements

Yes, the two pictures add requirements. Eight new product items and two platform items are now in the register (chapter 30), nine existing items grow, and seven new problems are in the log (chapter 35).

New items

ID	ITEM	FREE	PLUS	PRO	WHEN	FROM
F53	Onboarding wizard: risk profile, strategies, preferences, broker link	Profile, preferences	+ read-only broker link	Same	Month 3	Screen 1
F54	Personal preferences: focus sectors, excluded stocks, horizons, tighten-only risk presets, lower max per signal	Sectors, exclusions	+ horizons, max per signal	+ custom presets	Month 3	Screen 10
F55	Watchlist	10 stocks	50 + alerts	Unlimited	Month 2	Navigation
F56	Signal explorer: setup filters (momentum, breakout, pullback, oversold), cap and sector filters, risk label	Large cap	All	+ saved screens	Month 2	Screen 3
F57	Similar past setups table with outcomes and sample size	Count only	Top 3	Full list	Month 2	Screen 4
F58	Order hand-off to Kite: a pre-filled basket the user confirms at Zerodha	Paper only	Yes	+ order-list export	Month 4	Screen 5
F59	Insider, promoter and pledge disclosures feed (also a landmine input)	—	Flag	Full detail	Month 3	Brain inputs
F60	Public stock pages for search engines	Yes	Yes	Yes	Months 7–12	Architecture
C15	AI (LLM) use policy: allowed uses, banned uses, citations, cost caps, evaluation sets	Internal			Month 5	AI panel
C16	Execution quality: slippage and fill metrics	Internal (owner book)			Month 4	Outputs

Existing items that grow

ITEM	ADDITION
F47 Stock page	Tabs for overview, technical, fundamental, news and AI analysis; chart timeframes; key reasons; tablet layout
F50 Trade tracking	Expected path drawn as a band; expected-vs-actual chart
F51 Exit scenarios	Best / base / worst scenario card for new signals too
F10 Why not buy?	“Alternatives considered” on each decision

ITEM	ADDITION
F20 AI copilot	Stock comparison; phrasing rules (no “is a buy”, no return promises)
F37 Reports	Benchmark and tax tabs, monthly performance, equity curve; verified numbers only
F49 Sign-in	Microsoft sign-in recorded as “not now”
A1 Regime	ML regime model allowed only as a shadow challenger
F39 Lessons loop	Never changes risk parameters automatically

Decisions I need from you

- 1. Keep the engine's -4% stop, +8% target and up-to-15-day horizon in all designs until F41 proves otherwise? (P-021)
- 2. Users hand orders to Kite (basket) rather than placing them from TradeMind? (P-022)
- 3. Stay on DigitalOcean and treat the AWS diagram as a later reference? (P-024)
- 4. Skip Microsoft sign-in for now? (P-025)